

V15 Integrated Validity and Package Audit

Future Internet submission candidate

Status: PASS. All 61 integrated checks passed.

Scope

The audit preserves the V14 result hierarchy and verifies the separately locked seven-day BurstGPT robustness replication, trace provenance, numerical claims, manuscript compliance, archived code, LaTeX compilation, and PDF preflight. The original V11 strict result is not replaced or reinterpreted.

Verified result hierarchy

Study component	Audited result
V11 strict absolute gate	0/120 method runs certified
V11 retrospective noninferiority	DR-BGS 50/60; guarded GP 33/60
V12 exhaustive original regime	0/2,280 policy–scenario pairs certified
V13 compact-slice low-load confirmation	31 cross-method-distinct pairs passed both batches
V15 seven-day trace basis	19,431 rows; 17,896 positive-token rows
V15 chronological pools	11,810 search; 6,086 certification records
V15 seven-day high-load audit	0/2,280 policy–scenario pairs certified
V15 high-load nominations	DR-BGS 0/60; guarded GP 0/60; exhaustive 0/12
V15 low-load method-specific pairs	DR-BGS 6/6; guarded GP 12/12; exhaustive 6/6
V15 cross-method union	18/18 distinct pairs passed both batches
V15 maximum upper risk bound	0.00306935, below the 0.10 budget

Trace-provenance checks

The archived compact trace is an exact row-for-row prefix of the official BurstGPT release. The seven-day robustness window is defined by elapsed timestamps from 0 to 604,800 seconds rather than by a round record count. The official source hash, seven-day subset hash, protocol, configuration, execution code, and pre-result lock are archived and verified. The protocol lock precedes successful result generation.

Claim-boundary checks

The manuscript retains universal abstention at the original load levels, labels the operating-envelope work as exploratory, and describes the low-load result as a confirmation that the gate can issue positive decisions inside a feasible regime. It does not claim production safety, practical suitability of $\rho = 0.05$, DR-BGS superiority in the low-load confirmation, or new real-workload information from repeated bootstrap simulation.

Manuscript and technical checks

- The abstract contains 191 words.
- Author Contributions, Funding, Institutional Review Board Statement, Informed Consent Statement, Data Availability Statement, Acknowledgments, and Conflicts of Interest are present.
- All archived Python sources compile.
- The V15 LaTeX source compiles without undefined references, undefined citations, or overfull boxes.
- The 37-page PDF passed font and page preflight and visual inspection of the abstract, trace-provenance text, seven-day methods and results, combined figure, discussion, conclusion, and back matter.

External release boundary

The scientific package is internally complete. Before submission, reserve and insert the version-specific Zenodo DOI, publish and verify the immutable GitHub tag `v1.5.0`, then rerun the URL/DOI check on the final public files. Future Internet permits free-format initial submission; transfer to MDPI's current class package may therefore be completed either before submission or during production, but any conversion must be followed by another compile and visual audit.